

Introduction to ICT for Scientists

Science

May, 2026



Introduction to ICT for Scientists (A01781)

Short Title: Intro to ICT for Scientists
Faculty Science and Computing
Department Science
Cluster IT
Author JFRISBY
Credits 5

Level: Introductory

Description of Module / Aims

This module will provide the student with a general introduction to the fundamentals of PC-based computing from a scientific perspective. The student will gain a competency in the Microsoft Windows operating environment, as well as a proficiency in file and data management. An introduction to the basic principles of word processing, spreadsheets and presentations will be given. This will involve the manipulation of specific examples of a scientific nature, including laboratory results and data. Report writing and presentation skills will be acquired, such that the student can competently compile scientific data into a format acceptable for reports and presentations.

Programmes

COMP-0001	BSc (Common Entry) (WD_SSCIE_D)	1 / 1 / M
COMP-0001	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	1 / 1 / M
COMP-0001	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 1 / M
COMP-0001	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 1 / M
COMP-0001	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	1 / 1 / M
COMP-0001	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	1 / 1 / M
COMP-0001	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 1 / M
COMP-0001	BSc in Food Science with Business (WD_SFDSC_D)	1 / 1 / M
COMP-0001	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	1 / 1 / M
COMP-0001	BSc in Pharmaceutical Science (WD_SPHSC_D)	1 / 1 / M

Indicative Content

- Introduction to basic PC hardware components; introduction to Moodle; e-mail for attaching/ extracting files and e-mail etiquette; managing your online identity
- Familiarity with the MS Windows environment, particularly the use of Windows Explorer for the management of data files and folders
- Introduction to the use of MS Word for word processing & familiarity with basic formatting techniques
- Use of MS Word to format a document to a scientific report standard; generation of tables of scientific data
- Use of MS Equation Editor to construct mathematical equations
- Introduction to the use of MS PowerPoint to give presentations on a scientific topic
- Use of MS Excel to perform basic scientific calculations; use of Chart Wizard to graph simple functions
- Formatting of graphs & tables in MS Excel for incorporation into reports

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify PC components and gain a competency in basic computer housekeeping skills.
2. Apply MS Word and PowerPoint to compile and present scientific data in a clear and comprehensible manner, to a standard acceptable for scientific reports and presentations.
3. Apply Ms Excel to manipulate and graphically display trends in laboratory data.
4. Use Ms Excel to interpret data from scientific experiments.

Learning and Teaching Methods

- IT-based practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Participation</i>	10%	1,2,3,4
<i>In-Class Assessment</i>	40%	2
<i>In-Class Assessment</i>	50%	3,4

Assessment Criteria

- <40%:** Little or no understanding of the basic file management or applications of ICT to scientific analysis. Little or no demonstration of word processing, spreadsheet & presentation skills.
- 40%-49%:** Adequate file management skills & fair understanding of the basic applications of ICT to scientific analysis. Demonstration of rudimentary word processing, spreadsheet & presentation skills.
- 50%-59%:** As well as the above, will demonstrate good file management skills & a good understanding of some of the applications of ICT to scientific analysis. Will possess good word processing, spreadsheet & presentation skills.
- 60%-69%:** In addition, will demonstrate very good file management skills & understanding of the applications of ICT to scientific analysis. Will possess very good word processing, spreadsheet & presentation skills.
- 70%-100%:** All the above to an excellent level, in addition to a superior standard of word processing, spreadsheet & presentation skills.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Practical	36	
Independent Learning	99	

Essential Material

- "Introduction to ICT for Scientists module Moodle page." <http://www.moodle.wit.ie>

Supplementary Material

- Johnson, S. _Microsoft Office 2010 on Demand_. London: Pearson Education, 2010.

Requested Resources

- Room Type: Computer Lab